

Labor Supply

JEB104 Microeconomics I
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Basic Assumptions

Assumptions:

- Consumer consumes leisure R (useful and scarce good) and good C
- Consumption good C with price p_C
- Consumer endowed with *non-labor* income m
- Consumer endowed with $\bar{R}$ hours – used for leisure R or labor L
- Labor sold for wage w

Previously

- Consumer has initial endowment that determines his income (last time).

Now: Consumer could work to get income

- We apply the idea of endowment to analyze consumer's labor supply decision.
- Consumer is endowed with some leisure which he can sell (transform to labor) and get wage.
- Leisure is a scarce (limited) and useful good (necessary for relaxing, but also for consumption of other goods – e.g. to go to cinema you need 2 hours of your leisure).

Note: Labor **supply** – you supply your labor, firms demand it.Consumption good C

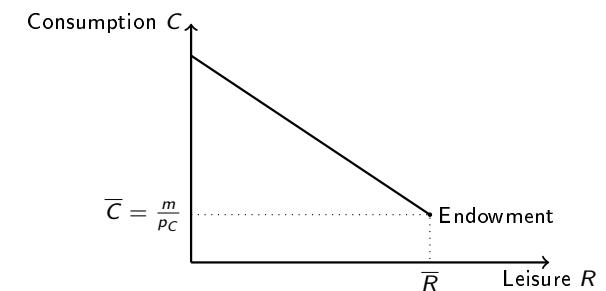
- Composite commodity representing consumption, one unit = 1 dollar spent on consumption, p_C price level, we will often assume that $p_C = 1$. In general model, we, however, keep p_C .

Non-labor income m and endowment $\bar{R}$

- non-labor income is your income which you get no matter whether you work or not (e.g. you could get it while renting your apartment to somebody else, or you could get monthly allowance from your parents to cover your costs while studying)
- spending your non-labor income m gives you consumption $\bar{C} = \frac{m}{p_C}$
- Initial endowment $\omega = (\bar{C}, \bar{R})$

Price of labor is wage w

Note that budget line only goes “left” – consumer cannot have more leisure than the upper limit



Budget Constraint

Budget constraint:

$$\begin{aligned}
 p_C C &\leq m + wL \\
 p_C C &\leq m + w(\bar{R} - R) \\
 p_C C + wR &\leq m + w\bar{R} \\
 \text{If } \bar{C} = \frac{m}{p_C} : \quad p_C C + wR &\leq p_C \bar{C} + w\bar{R}
 \end{aligned}$$

Value of consumption + leisure $\leq$ value of endowment (consumption + time)

Wage = opportunity cost of leisure

Budget constraint

- Consumption costs $p_C \cdot C$
- We pay our consumption from income we have – generated by working $w \cdot L$ (wage times amount of labor we supply) + non-labor income m
- Amount of labour we supply is exactly amount of leisure available $\bar{R}$ minus amount of leisure we consume R

$$p_C C + wR \leq m + w\bar{R}$$

- $m + w\bar{R}$ is wealth = value of endowment
- $w\bar{R}$ – income we can get when “selling” our entire leisure (means working all the time and not consuming any leisure)

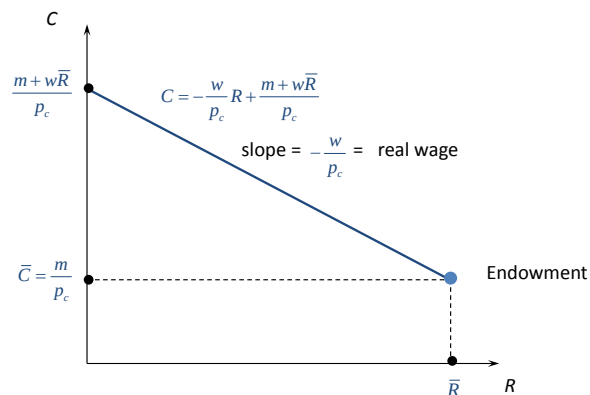
$$p_C C + wR \leq p_C \bar{C} + w\bar{R}$$

- This form of the BC is equivalent to the budget constraint discussed in the previous lecture: $\omega = (\bar{C}, \bar{R})$, $p_1 x_1 + p_2 x_2 \leq p_1 \omega_1 + p_2 \omega_2$

Wage = opportunity cost of leisure

- You decide to consume 1 extra hour of leisure – it costs you the wage you would receive if you worked.
- Therefore the slope of BC is $-\frac{w}{p_C}$ (price ratio, equivalent to $-\frac{p_1}{p_2}$ in the standard analysis)

Budget Constraint

Rewriting $p_C C + wR = m + w\bar{R}$ gives the function of the budget line:

$$C = -\frac{w}{p_C} R + \frac{m + w\bar{R}}{p_C}$$

Extremes:

- No work: $x = (\bar{R}, \bar{C})$, consumer cannot consume more leisure than $\bar{R}$ and more consumption than $\bar{C}$.
- No leisure: $x = (0, \frac{m + w\bar{R}}{p_C})$, consumer uses the non-labor income m plus what he gets by selling all leisure $w\bar{R}$ for consumption.

Real wage $\frac{w}{p_C}$ expresses how many units of consumption good can be bought from the wage w .

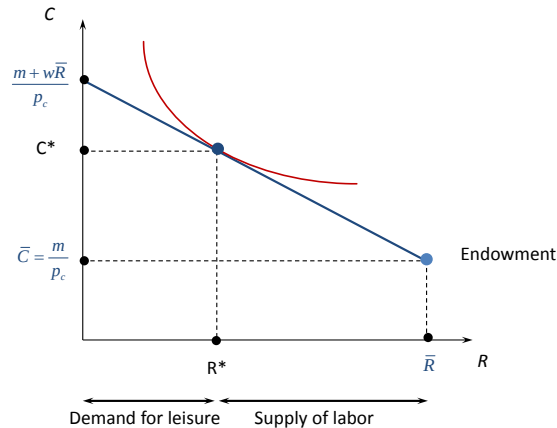
Optimum

Utility function:

$$u(C, R) = u(C, \bar{R} - L), \quad MU_R > 0$$

Interior optimum:

$$MRS_{CR} = -\frac{MU_R(R^*, C^*)}{MU_C(R^*, C^*)} = -\frac{w}{p_C}$$



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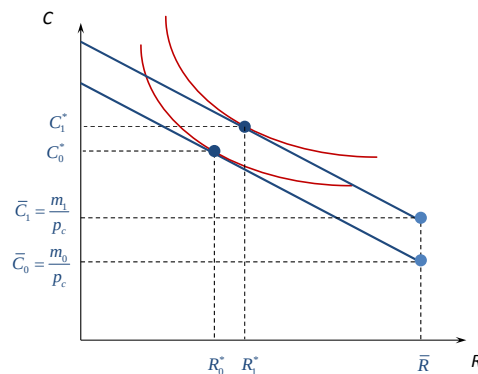
Labor Supply

Baseline Analysis
Comparative Statics

Change in Non-Labor Income
Change in Real Wage

Change in Non-Labor Income

Increase in m : $m_0 \rightarrow m_1$



Consumption: normal good

Leisure: normal good

- $\uparrow m \Rightarrow \uparrow C, \uparrow R$
- $\downarrow m \Rightarrow \downarrow C, \downarrow R$

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Labor Supply

We have already learned how the budget constraint looks like. Now we introduce a utility function $u(C, R)$. We usually assume that both consumption and leisure brings our consumer positive satisfaction, so $MU_C > 0$ and $MU_R > 0$.

In our model, $MU_R = -MU_L$, because $\frac{\partial u}{\partial L} = \frac{\partial u}{\partial R} \frac{\partial R}{\partial L} = MU_R \cdot (-1)$

This assumption is restricting the analysis, because some people like their work and working itself brings them satisfaction. Our theoretical consumer/worker, does not like working, one extra hour of work decreases his level of utility by the amount he could get while consuming one extra hour of leisure. So why does he work? Because he earns labor income which is spent on his consumption, and hence increases his utility level.

Labor $L^* = \bar{R} - R^*$, wage from working $wL^* = w(\bar{R} - R^*)$ is transformed to consumption $C^* - \bar{C}$, i.e. $C^* - \bar{C} = \frac{1}{p_C} w(\bar{R} - R^*)$.

In interior optimum, the optimality condition holds, i.e. MRS is equal to price ratio.

There can be corner solutions.

- In such case there will either be $R = \bar{R}$ or $R = 0$.
- If $|MRS_{CR}| > |\frac{w}{p_C}|$, then $R^* = \bar{R}$, i.e. the last consumed unit of R brings our consumer higher utility than what would additional consumption bring while working this one unit. Indifference curves are steeper than the budget line for all $R \in (0, \bar{R})$.
- If $|MRS_{CR}| < |\frac{w}{p_C}|$, then $R^* = 0$, i.e. the first unit of R would bring our consumer lower utility than what would additional consumption bring while working this last one unit. Indifference curves are flatter than the budget line for all $R \in (0, \bar{R})$.

From optimization, we can derive demand for leisure (as a function of real wage and non-labor income) and also demand for consumption (could be written as a function of $\frac{p_C}{w}$): $D_R(\frac{w}{p_C}, m)$, $D_C(\frac{w}{p_C}, m) = D_C(\frac{p_C}{w}, m)$

Labor supply is derived from the demand for leisure: $L(\frac{w}{p_C}, m) = \bar{R} - D_R(\frac{w}{p_C}, m)$

Let's discuss how changes in non-labor income and real wage affect optimum.

Increase in non-labor income (from m_0 to m_1) shifts the budget line up by $\frac{m_1 - m_0}{p_C}$.

We assume that consumption is a normal good.

- Generally, there are no substitutes to consumption (nothing better we can consume).
- This does not mean there are no substitute goods, but overall composite good is normal.

Leisure is also a normal good: 3 main reasons

1. Empirical: It is observed from the data.
2. Technical: Higher consumption requires more time, so if your income rises, you need more leisure (consumption is normal).
3. Psychological: People like free time, so with increase in income they increase consumption, but also free time (you do not need to desperately consume more and more).

Change in Real Wage

Increase in real wage:

$$\Delta \frac{w}{p_c} > 0$$

- Consumer is not worse off after the price change.
- If he works before the price change, he will be better off after the price change.

Decrease in real wage:

$$\Delta \frac{w}{p_c} < 0$$

- Consumer is not better off after the price change.
- If he works before the price change, he will be worse off after the price change.

Change in Demand for Consumption with Change in Real Wage

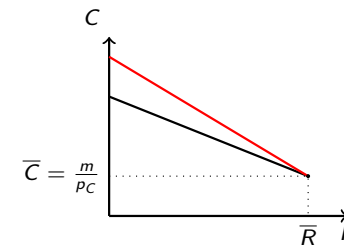
If real wage increases, $\Delta \frac{w}{p_c} > 0$:

- Consumption becomes relatively cheaper (you can get original consumption with less work) $\Rightarrow$ substitution effect $\uparrow C$
- Consumer earns higher labor income $\Rightarrow \uparrow$ real income $\Rightarrow$ income effect $\uparrow C$ because C is a normal good

Slutsky equation for cross-price change (the only difference is that we use real wage):

$$\underbrace{\frac{\partial D_C(\frac{p_c}{w}, m)}{\partial (\frac{w}{p_c})}}_{>0} = \underbrace{\frac{\partial h_C(\frac{p_c}{w}, u)}{\partial (\frac{w}{p_c})}}_{>0} \underbrace{-(D_R^* - \bar{R})}_{\geq 0} \underbrace{\frac{\partial D_C(\frac{p_c}{w}, m)}{\partial m}}_{>0}$$

Increase in real wage (new budget line steeper):



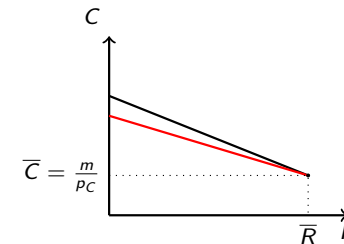
- Not working initially, $R^* = \bar{R}$:
 - Not working even after the price change means no change

in utility (steep ICs)

- If consumer starts working, he has to be better off (his initial optimum is still available).

- Working initially, $R^* < \bar{R}$:
 - From revealed preferences – initial optimum is available, but not chosen, so he must be better off.
 - Consumer is seller of leisure – increase of its price makes seller better off.

Decrease in real wage (new budget line flatter):



- Not working initially, $R^* = \bar{R}$:
 - Consumer will not start

working (would not make any sense), so no change in utility.

- Working initially, $R^* < \bar{R}$:
 - From revealed preferences – initial optimum is not affordable anymore, current optimum was available before the price change, so he must be worse off now.
 - Consumer is seller of leisure – decrease of its price makes seller worse off.

The analysis of whether consumer works more or less with change in real wage will follow.

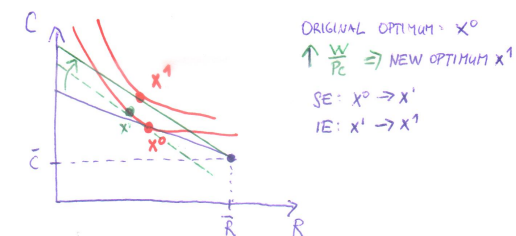
Firstly, we will study how change in real wage affects demand for consumption.

Hicksian demand – decreasing for own relative price $\frac{p_c}{w}$:

$$\frac{\partial h_C(\frac{p_c}{w}, u)}{\partial (\frac{p_c}{w})} < 0 \Rightarrow \frac{\partial h_C(\frac{p_c}{w}, u)}{\partial (\frac{w}{p_c})} = \frac{\partial h_C(\frac{p_c}{w}, u)}{\partial (\frac{p_c}{w})} \underbrace{\frac{\partial (\frac{p_c}{w})}{\partial (\frac{w}{p_c})}}_{-1} > 0$$

Real income increases: $m_{real} = L \cdot \frac{w}{p_c} \Rightarrow \frac{\partial m_{real}}{\partial (\frac{w}{p_c})} = L \cdot \bar{R} - D_R^* > 0$

More consumer works, more he is affected by the change in real wage.

Sensitivity to income: (C is a normal good) $\frac{\partial D_C(\frac{p_c}{w}, m)}{\partial m} > 0$ 

Change in Demand for Leisure with Change in Real Wage

If real wage increases, $\Delta \frac{w}{p_c} > 0$:

- Opportunity cost of leisure $\uparrow \Rightarrow$ more expensive not to work $\Rightarrow$ substitution effect $\downarrow R$
- Higher labor income $\Rightarrow \uparrow$ real income $\Rightarrow$ income effect $\uparrow R$
- Effects have opposite direction $\Rightarrow$ the total effect on demand for leisure is unclear

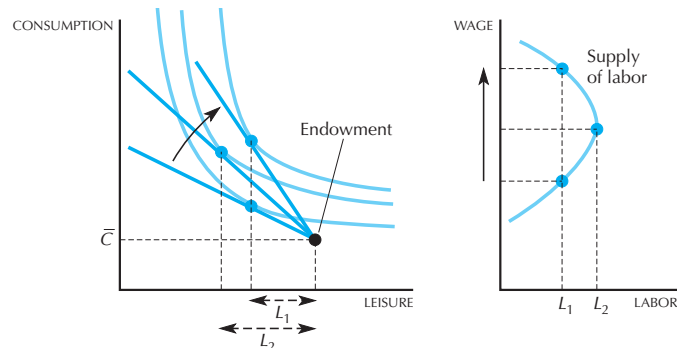
Slutsky equation for own-price change:

$$\frac{\partial D_R(\frac{w}{p_c}, m)}{\partial(\frac{w}{p_c})} = \underbrace{\frac{\partial h_R(\frac{w}{p_c}, u)}{\partial(\frac{w}{p_c})}}_{<0} \underbrace{\left[\underbrace{\frac{\partial D_R(\frac{w}{p_c}, m)}{\partial m}}_{>0} \right]}_{\geq 0} \underbrace{\frac{\partial D_R(\frac{w}{p_c}, m)}{\partial m}}_{>0}$$

$$\frac{\partial D_R(\frac{w}{p_c}, m)}{\partial(\frac{w}{p_c})} \leq 0 \iff |SE| \geq |EE + IE|$$

- For low $\frac{w}{p_c}$ (D_R^* close to $\bar{R}$): $|SE|$ tends to dominate $|EE + IE| \Rightarrow$ with $\uparrow \frac{w}{p_c}$ supply of labor increases and demand for leisure decreases
- For some higher $\frac{w}{p_c}$ (lower level D_R^*): $|SE|$ begins to be dominated by $|EE + IE| \Rightarrow$ with $\uparrow \frac{w}{p_c}$ supply of labor $\downarrow$ and demand for leisure $\uparrow$

Backward Bending Labor Supply



Low real wage:

- Utility from consumption purchased from the additional income is higher than disutility of working additional hour (and earning this additional income).
- Consumer works more with increase in real wage.

High real wage:

- Disutility of working additional hour dominates utility from consumption purchased from the additional income earned (lower as there is plenty of consumption).
- Consumer works less with increase in real wage.

Here, we analyze how demand for leisure (e.g. also supply of labor) changes with change in real wage.

Opportunity cost of leisure - it becomes less valuable to enjoy free time, not to work is more costly with wage increase

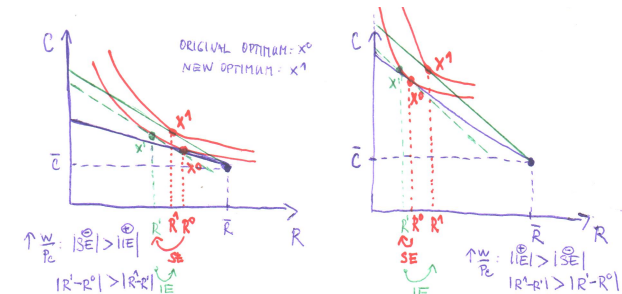
$$\frac{\partial h_R(\frac{w}{p_c}, u)}{\partial(\frac{w}{p_c})} < 0$$

Real income increases: $m_{real} = L \cdot \frac{w}{p_c} \Rightarrow \frac{\partial m_{real}}{\partial(\frac{w}{p_c})} = L^* = \bar{R} - D_R^* > 0$

Leisure is a normal good: $\frac{\partial D_R(\frac{w}{p_c}, m)}{\partial m} > 0$

Low $\frac{w}{p_c}$: consumer does not work much for low wage, so "total" income effect (EE + IE) is small (because D_R^* is close to $\bar{R}$). In addition, due to the fact that the consumer consumes a lot of leisure, it is not very scarce, and the consumer is willing to give up quite a lot of leisure to get additional consumption (substitution effect). If real wage increases, SE dominates, the fact that not working is now much more costly is more important than the fact that the consumer earns a little bit higher income (he works only a few hours). Increase in $\frac{w}{p_c}$ leads to increase of labor supply.

High $\frac{w}{p_c}$: consumer works a lot for high wage, so "total" income effect (EE + IE) is large (because D_R^* is far from $\bar{R}$). If real wage increases, labor income increases a lot because the consumer already works many hours (income effect). Due to the fact that the consumer does not consume much leisure (works a lot), leisure is scarcer and he is not willing to give up much leisure for additional consumption (so substitution effect is small). Hence, SE is dominated by (EE + IE) and increase in $\frac{w}{p_c}$ decreases supply of labor.



These effects would lead to backward-bending labor supply.

With low (real) wage, it pays off to work more with increase in wage to increase consumption (consumption is lower initially)

With a substantially high (real) wage, you rather enjoy free time (you already have enough consumption and therefore you do not need to increase it as much)